1) ~~784~~ # MACs = i/p × ~~o/p~~ neurons

Layer 1 $784 \rightarrow 256$:

$784 \times 256 = 200,704$

Layer 2 :

$256 \times 128 = 32,768$

Layer 3 :

$128 \times 10 = 1280$

2) ~~Layer~~ : Total MAC $= 234,752$ ✓
for 1 forward pass

~~= 939,008 bytes.~~

3) Total ~~# of trainable parameters~~ weight memory in bytes.

= total MAC × 4 bytes each

$= 234,752 \times 4$

$= 939,008$ bytes.

3) total # of trainable parameters

$= (784 \times 256) + (256 \times 128) + (128 \times 10)$

$= 234,752.$

5) Total activation memory

$= (784 + 256 + 128 + 10) \times 4$

$= 1178 \times 4$

$= 4712$ bytes. ✓

6) Arithmatic intensity

$= (2 \times \text{total MAC}) \,/\, (\text{weight bytes} + \text{activation bytes})$

$$= \frac{2 \times 234,752}{939,008 + 4712}$$

$$= \frac{469\,504}{943,720}$$

$$= 0.4975$$